

1 Adversarial robustness of online leverage-score sampling

Leverage-score sampling is a standard route to spectral approximation: keeping each row m_t of $A = \sum_t m_t m_t^\top$ with probability proportional to its leverage score $m_t^\top A^{-1} m_t$ yields a $(1 \pm \varepsilon)$ spectral approximation from only $\tilde{O}(d/\varepsilon^2)$ rows [9]. *Online* leverage-score sampling [3] makes these keep/discard decisions irrevocably as the rows arrive—scoring m_t against the matrix seen so far—and over a stream retains $\tilde{O}(d \text{ polylog } \kappa/\varepsilon^2)$ rows, where κ is the condition number.

This guarantee assumes an *oblivious* stream fixed in advance. Against an *adaptive* adversary, whose rows may depend on the algorithm’s past coin flips—as in a dynamic data structure—the bound degrades. Braverman, Hassidim, Matias, Schain, Silwal, and Zhou [1] first proved adaptive robustness for online leverage scores, but at a cost of $\tilde{O}(d^2 \kappa^2)$ rows; Jiang, Peng, and Weinstein [6] removed the polynomial dependence on κ , yet union-bounding over an ε -net of terminal directions still cost an extra factor d , giving $\tilde{O}(d^2 \text{ polylog } \kappa/\varepsilon^2)$. Whether this extra d is necessary remained open; a recent improvement of Kenneth-Mordoch and Sapir [7] closes the gap only in dimension $d = 1$.

The open question is resolved. Theorem 1.2 shows that $O(d \text{ polylog } \kappa/\varepsilon^2)$ rows suffice, and this matches the bound for oblivious adversary.

1.1 Setting and main result

Notation. Let d be the matrix dimension, $\preceq$ is the Löwner order, $\|\cdot\|_{\text{op}}$ is the operator (spectral) norm, tr is the trace, and for symmetric M we write e^M for the matrix exponential.

Definition 1.1 (Online leverage score sampling against adaptive adversary). The initial matrix $A_0 \succ 0$ is fixed. For $t = 1, \dots, T$:

- (i) an adaptive adversary reveals a row $m_t \in \mathbb{R}^d$, chosen as an arbitrary (deterministic or randomized) function of the entire past transcript $m_1, \xi_1, \dots, m_{t-1}, \xi_{t-1}$;
- (ii) the algorithm computes the probability $p_t \in (0, 1]$ with

$$p_t \geq \min\{\alpha \tau_t, 1\}, \quad \tau_t = m_t^\top \tilde{A}_{t-1}^{-1} m_t, \quad (1.1)$$

is the subsampled matrix.

- (iii) the algorithm draws a fresh coin $\xi_t \sim \text{Bernoulli}(p_t)$ and compute the new sub-sampled matrix:

$$\tilde{A}_t = A_0 + \sum_{s \leq t} \frac{\xi_s}{p_s} m_s m_s^\top \quad (1.2)$$

Finally, we assume conditioning bounds $\sigma_{\min}^2 I \preceq A_0$ and $A_T \preceq \sigma_{\max}^2 I$, and set $\kappa = \sigma_{\max}/\sigma_{\min}$.

We define A_t be the full matrix at step t , i.e.,

$$A_t = A_0 + \sum_{s=1}^t m_s m_s^\top \quad (1.3)$$

Let $\mathcal{F}_t$ be the σ -algebra generated by the transcript through round t , and let $\mathcal{F}_{t-1}^+ = \sigma(\mathcal{F}_{t-1}, m_t, p_t)$ be the information available just before the coin ξ_t is flipped; write $\mathbb{E}_{t-1}[\cdot] =$

$\mathbb{E}[\cdot \mid \mathcal{F}_{t-1}^+]$, the expectation over the single fresh coin ξ_t . Because A_t is monotone in t , the conditioning bounds give $\sigma_{\min}^2 I \preceq A_t \preceq \sigma_{\max}^2 I$ for all $t \leq T$. Quantities such as A_t , τ_t , p_t , $\tilde{A}_{t-1}$ are $\mathcal{F}_{t-1}^+$ -measurable; only ξ_t is random given $\mathcal{F}_{t-1}^+$.

Theorem 1.2. *Under setting 1.1, fix $\varepsilon \in (0, 1]$, set*

$$\alpha \geq C \cdot \varepsilon^{-2} (\log d + \log(1/\delta) + \log(\kappa)) \quad (1.4)$$

for some universal constant C , then with probability at least $1 - \delta$,

$$(1 - \varepsilon) A_t \preceq \tilde{A}_t \preceq (1 + \varepsilon) A_t \quad \text{simultaneously for all } t \leq T. \quad (1.5)$$

Section 1.2 states useful matrix inequalities, Section 1.3 proves Theorem 1.2; and Section 1.4 gives an upper bound on the sampled rows in Definition 1.1.

1.2 Preliminaries

We use four standard results: two trace inequalities, a maximal inequality for the final tail bound, and a martingale Bernstein bound for the row count.

Lemma 1.3 (Matrix MGF inequality [8, 10]). *Let H be a fixed symmetric matrix and X a random symmetric matrix. Then*

$$\mathbb{E} \operatorname{tr} e^{H+X} \leq \operatorname{tr} \exp(H + \log \mathbb{E} e^X),$$

The first inequality is the standard corollary of Lieb's concavity theorem [8]; see [10, §3]. The second holds because $A \preceq B$ gives $\lambda_i(H + A) \leq \lambda_i(H + B)$ for every i by Weyl's monotonicity theorem, so $\operatorname{tr} e^{H+A} = \sum_i e^{\lambda_i(H+A)} \leq \sum_i e^{\lambda_i(H+B)} = \operatorname{tr} e^{H+B}$.

Lemma 1.4 (Trace convexity under a contraction [2, 5]). *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be convex with $g(0) \leq 0$, let H be symmetric, and let Q be symmetric with $0 \preceq Q \preceq I$. Then*

$$\operatorname{tr} g(QHQ) \leq \operatorname{tr} (Q g(H) Q) = \operatorname{tr} (Q^2 g(H)).$$

In particular, applying this to $g(x) = e^x - 1$ (convex, with $g(0) = 0$) and rearranging,

$$\operatorname{tr} e^{QHQ} \leq \operatorname{tr} (Q^2 e^H) + \operatorname{tr} (I - Q^2). \quad (1.6)$$

The displayed inequality is the Brown–Kosaki trace inequality [2] (a symmetric Q with $0 \preceq Q \preceq I$ is a contraction, $\|Q\| \leq 1$); see also [5]. For (1.6), the inequality reads $\operatorname{tr}(e^{QHQ} - I) \leq \operatorname{tr}(Q^2(e^H - I))$, i.e. $\operatorname{tr} e^{QHQ} \leq \operatorname{tr}(Q^2 e^H) - \operatorname{tr}(Q^2) + \operatorname{tr} I$.

Lemma 1.5 (Ville's maximal inequality [11]). *If $(M_t)_{t \geq 0}$ is a nonnegative supermartingale, then for every $\lambda > 0$, $\mathbb{P}[\sup_{t \geq 0} M_t \geq \lambda] \leq \mathbb{E}[M_0]/\lambda$.*

Lemma 1.6 (Freedman's inequality [4]). *Let $(D_t)_{t \geq 1}$ be a martingale difference sequence adapted to a filtration $(\mathcal{F}_t)$, with $D_t \leq R$ for all t , and let $W = \sum_t \mathbb{E}[D_t^2 \mid \mathcal{F}_{t-1}]$ be its predictable quadratic variation. Then for all $u, v > 0$,*

$$\mathbb{P}\left[\sum_t D_t \geq u \text{ and } W \leq v\right] \leq \exp\left(-\frac{u^2}{2(v+Ru/3)}\right).$$

1.3 Proof of Theorem 1.2

Throughout the proof fix $\theta \in (0, \alpha/2]$, to be chosen at the end, and write $a = 2\theta^2/\alpha$. Define the *normalized error*

$$Z_t = A_t^{-1/2}(\tilde{A}_t - A_t)A_t^{-1/2}, \quad (1.7)$$

a symmetric $\mathcal{F}_t$ -measurable matrix with $Z_0 = 0$ (as $\tilde{A}_0 = A_0$). Conjugating (1.5) by $A_t^{1/2}$ shows that (1.5) at time t is equivalent to $\|Z_t\|_{\text{op}} \leq \varepsilon$. We control the trace-exponential potentials

$$\Phi_{\pm}(Z) = \text{tr } e^{\pm\theta Z}.$$

Lemma 1.7 (One-step recursion). *Set $v_t = A_t^{-1/2}m_t$, $P_t = v_t v_t^\top$, and $Q_t = (I - P_t)^{1/2}$. Then:*

(i) $A_t^{-1/2}A_{t-1}A_t^{-1/2} = I - P_t = Q_t^2$, and P_t is the rank-one projector scaled by the true ridge leverage score

$$\ell_t := \text{tr } P_t = m_t^\top A_t^{-1}m_t \in [0, 1]; \quad (1.8)$$

(ii) $A_t^{-1/2}A_{t-1}^{1/2} = Q_t U_t$ for an orthogonal matrix U_t ;

(iii) with $\eta_t = \frac{\xi_t}{p_t} - 1$ and $\bar{Z}_{t-1} = U_t Z_{t-1} U_t^\top$,

$$Z_t = Q_t \bar{Z}_{t-1} Q_t + \eta_t P_t. \quad (1.9)$$

Proof. (i). Since $A_{t-1} = A_t - m_t m_t^\top$, conjugating by $A_t^{-1/2}$ gives $A_t^{-1/2}A_{t-1}A_t^{-1/2} = I - v_t v_t^\top = I - P_t$. Here $\ell_t = v_t^\top v_t = m_t^\top A_t^{-1}m_t < 1$ because $A_t = A_{t-1} + m_t m_t^\top \succ m_t m_t^\top$, so the only nonzero eigenvalue of P_t is $\ell_t < 1$ and $I - P_t \succ 0$ equals Q_t^2 .

(ii). The matrix $C := A_t^{-1/2}A_{t-1}^{1/2}$ satisfies $CC^\top = A_t^{-1/2}A_{t-1}A_t^{-1/2} = Q_t^2$, so Q_t is its left polar factor and $C = Q_t U_t$ with $U_t = Q_t^{-1}C$ orthogonal ($U_t U_t^\top = Q_t^{-1}C C^\top Q_t^{-1} = I$).

(iii). By (1.3) and (1.2), $\tilde{A}_t - A_t = (\tilde{A}_{t-1} - A_{t-1}) + \eta_t m_t m_t^\top$. Conjugating by $A_t^{-1/2}$ and using $\tilde{A}_{t-1} - A_{t-1} = A_{t-1}^{1/2}Z_{t-1}A_{t-1}^{1/2}$,

$$Z_t = C Z_{t-1} C^\top + \eta_t v_t v_t^\top = Q_t U_t Z_{t-1} U_t^\top Q_t + \eta_t P_t = Q_t \bar{Z}_{t-1} Q_t + \eta_t P_t. \quad \square$$

Lemma 1.8 (Conditional matrix MGF). *Suppose the sketch is accurate at $t-1$, i.e. $(1-\varepsilon)A_{t-1} \preceq \tilde{A}_{t-1} \preceq (1+\varepsilon)A_{t-1}$. Write $P_t = \ell_t w_t w_t^\top$ with $\|w_t\|_2 = 1$. Then for every $\theta \in [0, \alpha/2]$ and either sign,*

$$\log \mathbb{E}_{t-1} e^{\pm\theta \eta_t P_t} \preceq \frac{2\theta^2}{\alpha} P_t. \quad (1.10)$$

Proof. If $p_t = 1$ then $\xi_t = 1$, $\eta_t = 0$, and the left side is zero; assume $p_t < 1$, so $\min\{\alpha\tau_t, 1\} = \alpha\tau_t$ and $p_t \geq \alpha\tau_t$.

Score comparison. The schedule uses the sketch score $\tau_t = m_t^\top \tilde{A}_{t-1}^{-1}m_t$, but the potential is normalized by the true A_t . From $\tilde{A}_{t-1} \preceq (1+\varepsilon)A_{t-1}$ we get $\tilde{A}_{t-1}^{-1} \succeq (1+\varepsilon)^{-1}A_{t-1}^{-1}$, and since $A_{t-1} \preceq A_t$,

$$\tau_t = m_t^\top \tilde{A}_{t-1}^{-1}m_t \geq \frac{1}{1+\varepsilon} m_t^\top A_{t-1}^{-1}m_t \geq \frac{1}{1+\varepsilon} m_t^\top A_t^{-1}m_t = \frac{\ell_t}{1+\varepsilon}.$$

Hence $\ell_t \leq (1 + \varepsilon)\tau_t$, and with $p_t \geq \alpha\tau_t$ and $\varepsilon \leq 1$,

$$\frac{\ell_t}{p_t} \leq \frac{\ell_t}{\alpha\tau_t} \leq \frac{1 + \varepsilon}{\alpha} \leq \frac{2}{\alpha}. \quad (1.11)$$

Set $Y_t = \eta_t \ell_t$. Now we control the LHS expectation in Eq. (1.10)

Boundedness. When $\xi_t = 0$, $|Y_t| = \ell_t \leq \ell_t/p_t$; when $\xi_t = 1$, $|Y_t| = \frac{1-p_t}{p_t}\ell_t \leq \ell_t/p_t$. So by (1.11) $|Y_t| \leq 2/\alpha$, and for $\theta \leq \alpha/2$ we have $\theta|Y_t| \leq 1$.

Mean and variance. As $\mathbb{E}_{t-1}\xi_t = p_t$, we have $\mathbb{E}_{t-1}\eta_t = 0$, so $\mathbb{E}_{t-1}Y_t = 0$, and using (1.11),

$$\mathbb{E}_{t-1}Y_t^2 = \ell_t^2 \mathbb{E}_{t-1}\eta_t^2 = \ell_t^2 \frac{1-p_t}{p_t} \leq \frac{\ell_t^2}{p_t} = \ell_t \cdot \frac{\ell_t}{p_t} \leq \frac{2\ell_t}{\alpha}.$$

Scalar MGF. Since $\theta|Y_t| \leq 1$, the elementary bound $e^x \leq 1 + x + x^2$ for $|x| \leq 1$ gives, for either sign,

$$\mathbb{E}_{t-1}e^{\pm\theta Y_t} \leq 1 + \theta\mathbb{E}_{t-1}(\pm Y_t) + \theta^2\mathbb{E}_{t-1}Y_t^2 \leq 1 + \frac{2\theta^2\ell_t}{\alpha} \leq \exp\left(\frac{2\theta^2\ell_t}{\alpha}\right).$$

Matrix MGF. Since $\eta_t P_t = Y_t w_t w_t^\top$ is supported on the rank-one projector $w_t w_t^\top$, we have $e^{\pm\theta\eta_t P_t} = I + (e^{\pm\theta Y_t} - 1)w_t w_t^\top$, hence $\mathbb{E}_{t-1}e^{\pm\theta\eta_t P_t} = I + (\mathbb{E}_{t-1}e^{\pm\theta Y_t} - 1)w_t w_t^\top$. Taking the matrix logarithm (a function of the single eigendirection w_t),

$$\log \mathbb{E}_{t-1}e^{\pm\theta\eta_t P_t} = \left(\log \mathbb{E}_{t-1}e^{\pm\theta Y_t}\right) w_t w_t^\top \preceq \frac{2\theta^2\ell_t}{\alpha} w_t w_t^\top = \frac{2\theta^2}{\alpha} P_t. \quad \square$$

Lemma 1.9 (One-step trace recursion). *With $a = 2\theta^2/\alpha$ and $\theta \in (0, \alpha/2]$, on the event $(1 - \varepsilon)A_{t-1} \preceq \tilde{A}_{t-1} \preceq (1 + \varepsilon)A_{t-1}$, for either sign,*

$$\mathbb{E}_{t-1} \Phi_\pm(Z_t) \leq \Phi_\pm(Z_{t-1}) + e^a \ell_t. \quad (1.12)$$

Proof. By (1.9), $\pm\theta Z_t = (\pm\theta Q_t \bar{Z}_{t-1} Q_t) + (\pm\theta\eta_t P_t)$. Write

$$H_t := \pm\theta Q_t \bar{Z}_{t-1} Q_t, \quad X_t := \pm\theta\eta_t P_t,$$

so that $\Phi_\pm(Z_t) = \text{tr } e^{H_t + X_t}$. Conditionally on $\mathcal{F}_{t-1}^+$ the matrix H_t is fixed—it depends only on A_{t-1}, A_t (through Q_t, U_t) and on Z_{t-1} —while X_t is random through ξ_t alone. We bound $\mathbb{E}_{t-1}\Phi_\pm(Z_t)$ in three steps.

Step 1 (matrix MGF subadditivity). Apply the first inequality of Lemma 1.3, conditionally on $\mathcal{F}_{t-1}^+$, with fixed matrix H_t and random matrix X_t :

$$\mathbb{E}_{t-1}\Phi_\pm(Z_t) = \mathbb{E}_{t-1} \text{tr } e^{H_t + X_t} \leq \text{tr } \exp\left(H_t + \log \mathbb{E}_{t-1}e^{X_t}\right).$$

Step 2 (moment bound). The conditional log-MGF of $X_t = \pm\theta\eta_t P_t$ is controlled by (1.10):

$$\log \mathbb{E}_{t-1}e^{X_t} \preceq \frac{2\theta^2}{\alpha} P_t = a P_t.$$

Step 3 (monotonicity). Since $\log \mathbb{E}_{t-1}e^{X_t} \preceq a P_t$ and H_t is fixed, the second (monotonicity) part of Lemma 1.3 substitutes the PSD upper bound back in:

$$\text{tr } \exp\left(H_t + \log \mathbb{E}_{t-1}e^{X_t}\right) \leq \text{tr } \exp(H_t + a P_t).$$

Chaining the three steps and unfolding H_t ,

$$\mathbb{E}_{t-1}\Phi_{\pm}(Z_t) \leq \text{tr} \exp\left(\pm \theta Q_t \bar{Z}_{t-1} Q_t + aP_t\right). \quad (1.13)$$

Write $H = \pm \theta \bar{Z}_{t-1} - aI$. Since $Q_t^2 = I - P_t$,

$$Q_t H Q_t + aI = \pm \theta Q_t \bar{Z}_{t-1} Q_t - a(I - P_t) + aI = \pm \theta Q_t \bar{Z}_{t-1} Q_t + aP_t,$$

so, as aI is scalar,

$$\text{tr} \exp\left(\pm \theta Q_t \bar{Z}_{t-1} Q_t + aP_t\right) = e^a \text{tr} e^{Q_t H Q_t}. \quad (1.14)$$

By the specialization (1.6) of Lemma 1.4,

$$e^a \text{tr} e^{Q_t H Q_t} \leq e^a \left[\text{tr} (Q_t^2 e^H) + \text{tr} (I - Q_t^2) \right]. \quad (1.15)$$

Here $\text{tr}(I - Q_t^2) = \text{tr} P_t = \ell_t$, contributing $e^a \ell_t$. For the other term, $e^H = e^{-a} e^{\pm \theta \bar{Z}_{t-1}}$, so

$$e^a \text{tr} (Q_t^2 e^H) = \text{tr} (Q_t^2 e^{\pm \theta \bar{Z}_{t-1}}) \leq \text{tr} e^{\pm \theta \bar{Z}_{t-1}} = \text{tr} e^{\pm \theta Z_{t-1}} = \Phi_{\pm}(Z_{t-1}), \quad (1.16)$$

where the inequality uses $0 \preceq Q_t^2 \preceq I$ with $e^{\pm \theta \bar{Z}_{t-1}} \succeq 0$, and the last equality uses unitary invariance of the trace ($\bar{Z}_{t-1} = U_t Z_{t-1} U_t^\top$). Combining Eq. (1.13)–(1.16) gives (1.12). $\square$

Lemma 1.10 (Deterministic leverage budget). *Pathwise*, $S_t := \sum_{s=1}^t \ell_s \leq \log \frac{\det A_t}{\det A_0} \leq 2d \log \kappa$

Proof. Since $A_{t-1} = A_t - m_t m_t^\top$, the matrix determinant lemma gives $\det A_{t-1} = \det A_t (1 - m_t^\top A_t^{-1} m_t) = \det A_t (1 - \ell_t)$, so $\ell_t = 1 - \frac{\det A_{t-1}}{\det A_t} \leq \log \frac{\det A_t}{\det A_{t-1}}$ (using $1 - x^{-1} \leq \log x$). Summing telescopes to $S_t \leq \log \frac{\det A_t}{\det A_0}$. Finally, the eigenvalues of A_t lie in $[\sigma_{\min}^2, \sigma_{\max}^2]$, so $\det A_t \leq \sigma_{\max}^{2d}$ and $\det A_0 \geq \sigma_{\min}^{2d}$, whence $\log \frac{\det A_t}{\det A_0} \leq 2d \log(\sigma_{\max}/\sigma_{\min}) = 2d \log \kappa$. $\square$

Proof of Theorem 1.2. Take $\theta = \alpha\epsilon/4$, so $\theta \leq \alpha/2$ and $a = 2\theta^2/\alpha = \alpha\epsilon^2/8$. Because the score comparison underlying Lemma 1.9 holds only while the sketch is accurate, we localize at the first failure time

$$\sigma = \inf\{t \geq 1 : \|Z_t\|_{\text{op}} \geq \epsilon\} \quad (\sigma = \infty \text{ if no failure}).$$

Since $\{\sigma \geq t\} = \{\|Z_s\|_{\text{op}} < \epsilon \text{ for all } s \leq t-1\}$ is $\mathcal{F}_{t-1}$ -measurable, σ is a stopping time. For either sign define

$$\mathcal{M}_t^{\pm} = \Phi_{\pm}(Z_t) + e^a (2d \log \kappa - S_t) \geq 0$$

(nonnegative since $S_t \leq 2d \log \kappa$ by Lemma 1.10 and $\Phi_{\pm} \geq 0$), and consider the stopped process $\mathcal{M}_{t \wedge \sigma}^{\pm}$.

The stopped process is a supermartingale. On $\{\sigma \leq t-1\}$ it is frozen, so the increment is 0. On $\{\sigma \geq t\}$ —an $\mathcal{F}_{t-1}$ -measurable event on which $\|Z_{t-1}\|_{\text{op}} < \epsilon$, i.e. $(1-\epsilon)A_{t-1} \preceq \tilde{A}_{t-1} \preceq (1+\epsilon)A_{t-1}$ —Lemma 1.9 applies, and since S_t is $\mathcal{F}_{t-1}^+$ -measurable,

$$\mathbb{E}_{t-1} \mathcal{M}_t^{\pm} \leq [\Phi_{\pm}(Z_{t-1}) + e^a \ell_t] + e^a (2d \log \kappa - S_{t-1} - \ell_t) = \mathcal{M}_{t-1}^{\pm}.$$

Taking $\mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$ (using $\mathcal{F}_{t-1} \subseteq \mathcal{F}_{t-1}^+$ and that $\{\sigma \geq t\}$ is $\mathcal{F}_{t-1}$ -measurable) yields $\mathbb{E}[\mathcal{M}_{t \wedge \sigma}^\pm \mid \mathcal{F}_{t-1}] \leq \mathcal{M}_{(t-1) \wedge \sigma}^\pm$. Thus $(\mathcal{M}_{t \wedge \sigma}^\pm)_{t \geq 0}$ is a nonnegative supermartingale with $\mathcal{M}_0^\pm = \Phi_\pm(Z_0) + e^a \cdot 2d \log \kappa = d + e^a \cdot 2d \log \kappa$ (using $Z_0 = 0$, $\text{tr } I = d$).

Tail bound. On $\{\sigma \leq T\}$ we have $\|Z_\sigma\|_{\text{op}} \geq \varepsilon$, so $\lambda_{\max}(\pm Z_\sigma) \geq \varepsilon$ for at least one sign, whence $\mathcal{M}_\sigma^\pm \geq \Phi_\pm(Z_\sigma) = \text{tr } e^{\pm \theta Z_\sigma} \geq e^{\theta \varepsilon}$ for that sign and $\sup_t \mathcal{M}_{t \wedge \sigma}^\pm \geq e^{\theta \varepsilon}$. By Ville's inequality (Lemma 1.5) applied to each sign and a union bound,

$$\mathbb{P}[\sigma \leq T] \leq \sum_{\pm} \mathbb{P}\left[\sup_t \mathcal{M}_{t \wedge \sigma}^\pm \geq e^{\theta \varepsilon}\right] \leq 2(d + e^a \cdot 2d \log \kappa) e^{-\theta \varepsilon}.$$

Since $\theta \varepsilon = \alpha \varepsilon^2 / 4 = 2a$, we have $(d + 2e^a d \log \kappa) e^{-2a} \leq 3d \log \kappa \cdot e^{-a}$, so $\mathbb{P}[\sigma \leq T] \leq 3d \log \kappa e^{-\alpha \varepsilon^2 / 8}$. Condition (1.4) makes this at most δ . $\square$

1.4 The number of sampled rows

Finally, we bound the total number of sampled rows, this is a well known result in the literature and we provide a short proof adapt to our parameter setting.

Corollary 1.11 (Number of sampled rows). *Use the canonical schedule $p_t = \min\{\alpha \tau_t, 1\}$ with α as in Theorem 1.2 and $\varepsilon \leq \frac{1}{2}$. With probability at least $1 - \delta$ the spectral guarantee (1.5) holds, and on that event*

$$\sum_{t=1}^T p_t \leq 4\alpha \sum_{t=1}^T \ell_t \leq 8\alpha d \log \kappa. \quad (1.17)$$

Consequently, for any $\rho \in (0, 1)$, with probability at least $1 - \delta - \rho$,

$$\sum_{t=1}^T \xi_t = O\left(\alpha d \log \kappa + \log \frac{1}{\rho}\right) = O\left(d \varepsilon^{-2} \log \kappa \log \frac{d \log(\kappa)}{\delta} + \log \frac{1}{\rho}\right). \quad (1.18)$$

Proof. Condition on the event (1.5) (probability $\geq 1 - \delta$ by Theorem 1.2); then $(1 - \varepsilon)A_{t-1} \preceq \tilde{A}_{t-1}$ for all t , so $\tau_t = m_t^\top \tilde{A}_{t-1}^{-1} m_t \leq \frac{1}{1 - \varepsilon} m_t^\top A_{t-1}^{-1} m_t = \frac{1}{1 - \varepsilon} \cdot \frac{\ell_t}{1 - \ell_t}$, using $m_t^\top A_{t-1}^{-1} m_t = \ell_t / (1 - \ell_t)$ (Sherman–Morrison). We claim $p_t \leq 4\alpha \ell_t$. If $\ell_t \leq \frac{1}{2}$ then $\frac{\ell_t}{1 - \ell_t} \leq 2\ell_t$, so $p_t \leq \alpha \tau_t \leq \frac{2\alpha}{1 - \varepsilon} \ell_t \leq 4\alpha \ell_t$ (as $\varepsilon \leq \frac{1}{2}$); if $\ell_t > \frac{1}{2}$ then $p_t \leq 1 < 2\ell_t \leq 4\alpha \ell_t$. Summing and applying Lemma 1.10 gives (1.17).

For the kept rows, write $\sum_t \xi_t = \sum_t p_t + \sum_t (\xi_t - p_t)$ and apply Lemma 1.6 to $D_t = \xi_t - p_t$ (so $|D_t| \leq 1$, hence $R = 1$), whose predictable quadratic variation is $W = \sum_t p_t(1 - p_t) \leq \sum_t p_t$. On the event (1.5), $W \leq 8\alpha d \log \kappa =: V$ by (1.17); taking $v = V$ and $u = V + \frac{10}{3} \log \frac{1}{\rho}$ makes the Freedman bound at most ρ (the exponent is $\geq \frac{3u}{8} \geq \log \frac{1}{\rho}$). Hence with probability $\geq 1 - \delta - \rho$, both (1.5) holds and $\sum_t \xi_t \leq 2V + \frac{10}{3} \log \frac{1}{\rho}$. Combining with $\alpha = O(\varepsilon^{-2} \log \frac{d \log(e\kappa)}{\delta})$ gives (1.18). $\square$

References

- [1] Vladimir Braverman, Avinatan Hassidim, Yossi Matias, Mariano Schain, Sandeep Silwal, and Samson Zhou. Adversarial robustness of streaming algorithms through importance sampling. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021. arXiv:2106.14952.

- [2] Lawrence G. Brown and Hideki Kosaki. Jensen’s inequality in semifinite von Neumann algebras. *Journal of Operator Theory*, 23(1):3–19, 1990.
- [3] Michael B. Cohen, Cameron Musco, and Jakub Pachocki. Online row sampling. In *Approximation, Randomization, and Combinatorial Optimization (APPROX/RANDOM)*, 2016.
- [4] David A. Freedman. On tail probabilities for martingales. *The Annals of Probability*, 3(1):100–118, 1975.
- [5] Frank Hansen and Gert K. Pedersen. Jensen’s operator inequality. *Bulletin of the London Mathematical Society*, 35(4):553–564, 2003.
- [6] Shunhua Jiang, Binghui Peng, and Omri Weinstein. The complexity of dynamic least-squares regression. In *64th IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, pages 1605–1627, 2023. arXiv:2201.00228.
- [7] Yotam Kenneth-Mordoch and Shay Sapir. On the adversarial robustness of online importance sampling, 2025.
- [8] Elliott H. Lieb. Convex trace functions and the Wigner–Yanase–Dyson conjecture. *Advances in Mathematics*, 11(3):267–288, 1973.
- [9] Daniel A. Spielman and Nikhil Srivastava. Graph sparsification by effective resistances. *SIAM Journal on Computing*, 40(6):1913–1926, 2011.
- [10] Joel A. Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012.
- [11] Jean Ville. *Étude critique de la notion de collectif*. Gauthier-Villars, Paris, 1939.